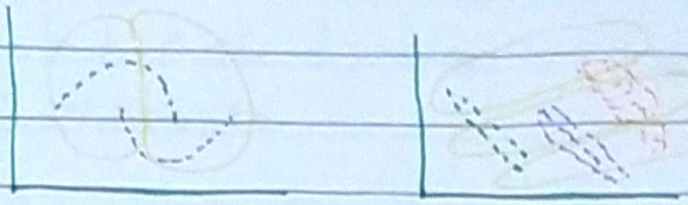
⇒ Distance to centriod that k-means uses won't be efficient in cases like This

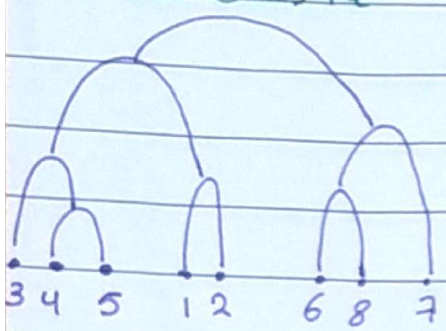


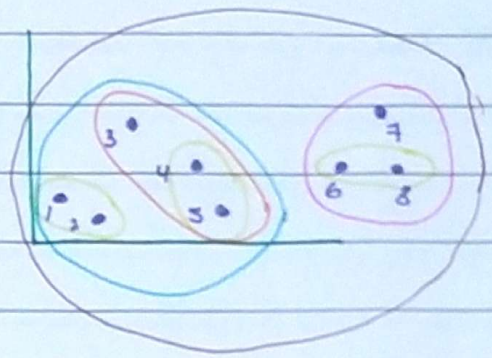
⇒ k-means will do a good Job in a larger number of Clusters

Hierical clustring agglomerative clustring

① single link clustring..

- assume each Point is a cluster
- calculate the distance between every 2 Points and the Points with the smalest distance to each other will be a cluster
- measure the distance between Point and the closest Point in the cluster





② Complete link clustring

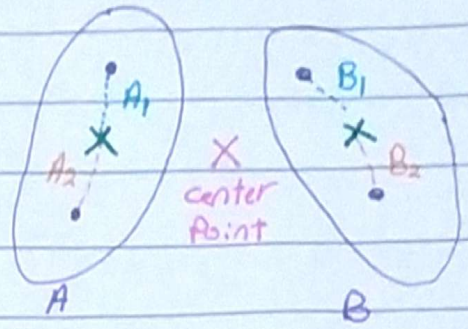
Same to the single link But the distance between Point and Clusters is the farthest distance between Points

3 Average link Clustring.

- same as single link but the distance is the average between every Point & other Points.

4 Ward's method : The defult Method in sklearn.

- distance between 2 Clusters $\Delta(A,B)$:



$$\Delta(A,B) = C_1^2 + C_2^2 + C_3^2 + C_4^2 - A_1^2 - A_2^2 - B_1^2 - B_2^2$$

⟹ DB Scan

- بنشوف كل نقطه والنقط اللى حواليها لو قريبين لبعض وعددهم >= أقل عدد من النقط اللى محددينه ، حينه يكون cluster كدا بنجمعهم فى cluster غير كدا بنعتبر النقطه دى Noise والنقطه اللى بيبقى حواليها باقى النقط دى بنسميها core Point